

SC3103: Embedded Programming

Tutorial 4: PCP/SRP Schedulability

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1. Consider a sporadic real-time task set comprising 10 tasks $\tau_1, \dots, \tau_{10}$, where τ_1 has the highest and τ_{10} has the lowest priority. Suppose there are 3 resources R_1, R_2 and R_3 , shared by these tasks. The usage profile for these resources are as follows ($RAT_{i,j}$ denotes maximum resource access time for resource R_j by task τ_i):

- $RAT_{3,1} = 5$
- $RAT_{5,1} = 3, RAT_{5,3} = 2$
- $RAT_{7,1} = 4, RAT_{7,2} = 5$
- $RAT_{8,2} = 2, RAT_{8,3} = 2$
- $RAT_{10,1} = 1, RAT_{10,2} = 2, RAT_{10,3} = 3$

Any $RAT_{i,j}$ not mentioned above by default has value 0 and it can be assumed that task τ_i does not use resource R_j . Compute the worst-case blocking times B_i for all the 10 tasks under PCP/SRP.

2. Consider a sporadic real-time task set comprising 7 tasks $\tau_1, \dots, \tau_7$, where τ_1 has the highest and τ_7 has the lowest priority. Suppose there are 3 resources R_1, R_2 and R_3 , shared by these tasks. The usage profile for these resources are as follows ($RAT_{i,j}$ denotes maximum resource access time for resource R_j by task τ_i):

- $RAT_{1,1} = 5$
- $RAT_{2,1} = 4, RAT_{2,2} = 3$
- $RAT_{3,1} = 4, RAT_{3,2} = 5$
- $RAT_{5,3} = 2$
- $RAT_{7,3} = 3$

Any $RAT_{i,j}$ not mentioned above by default has value 0 and it can be assumed that task τ_i does not use resource R_j . Compute the worst-case blocking times B_i for all the 7 tasks under PCP/SRP.